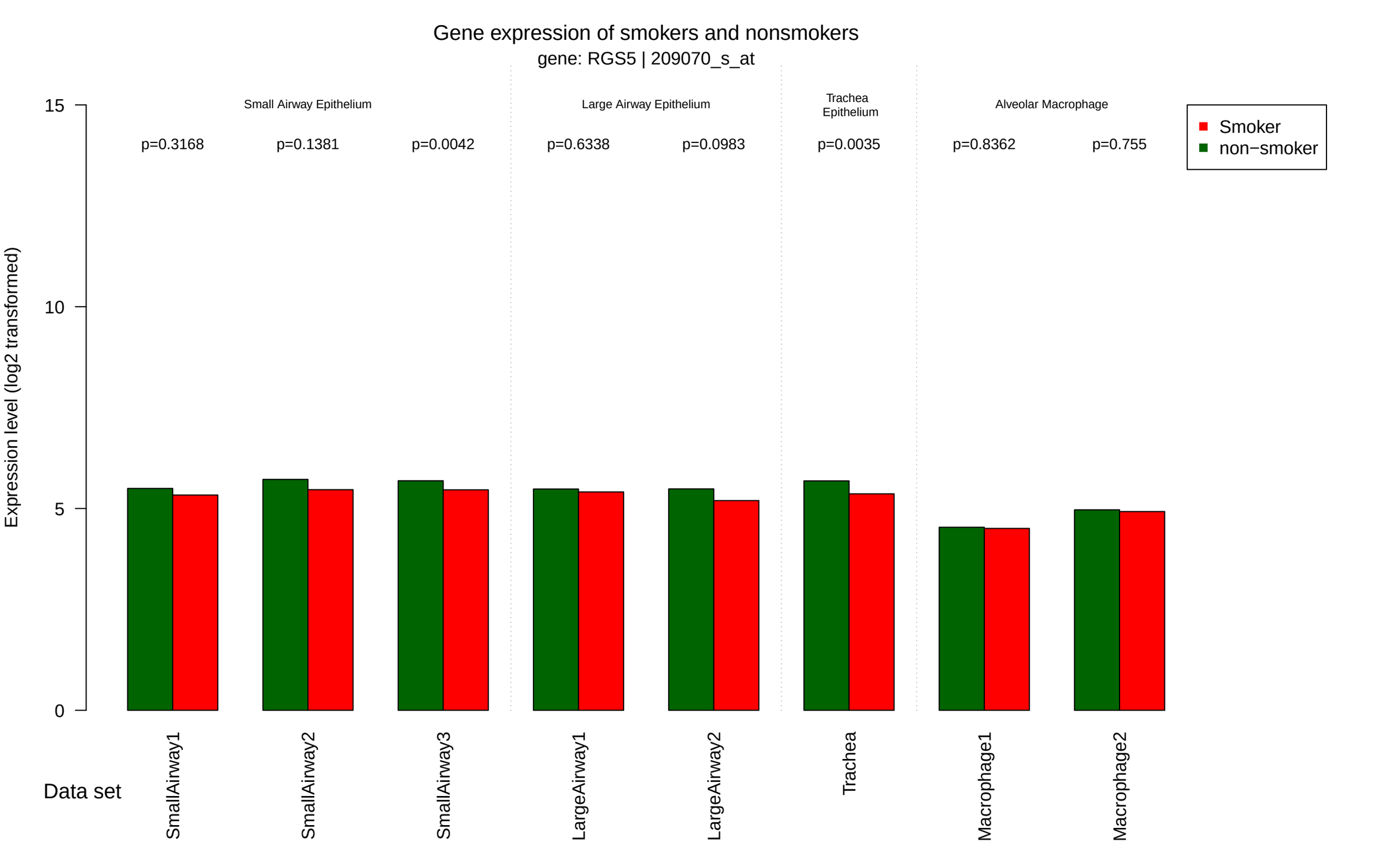
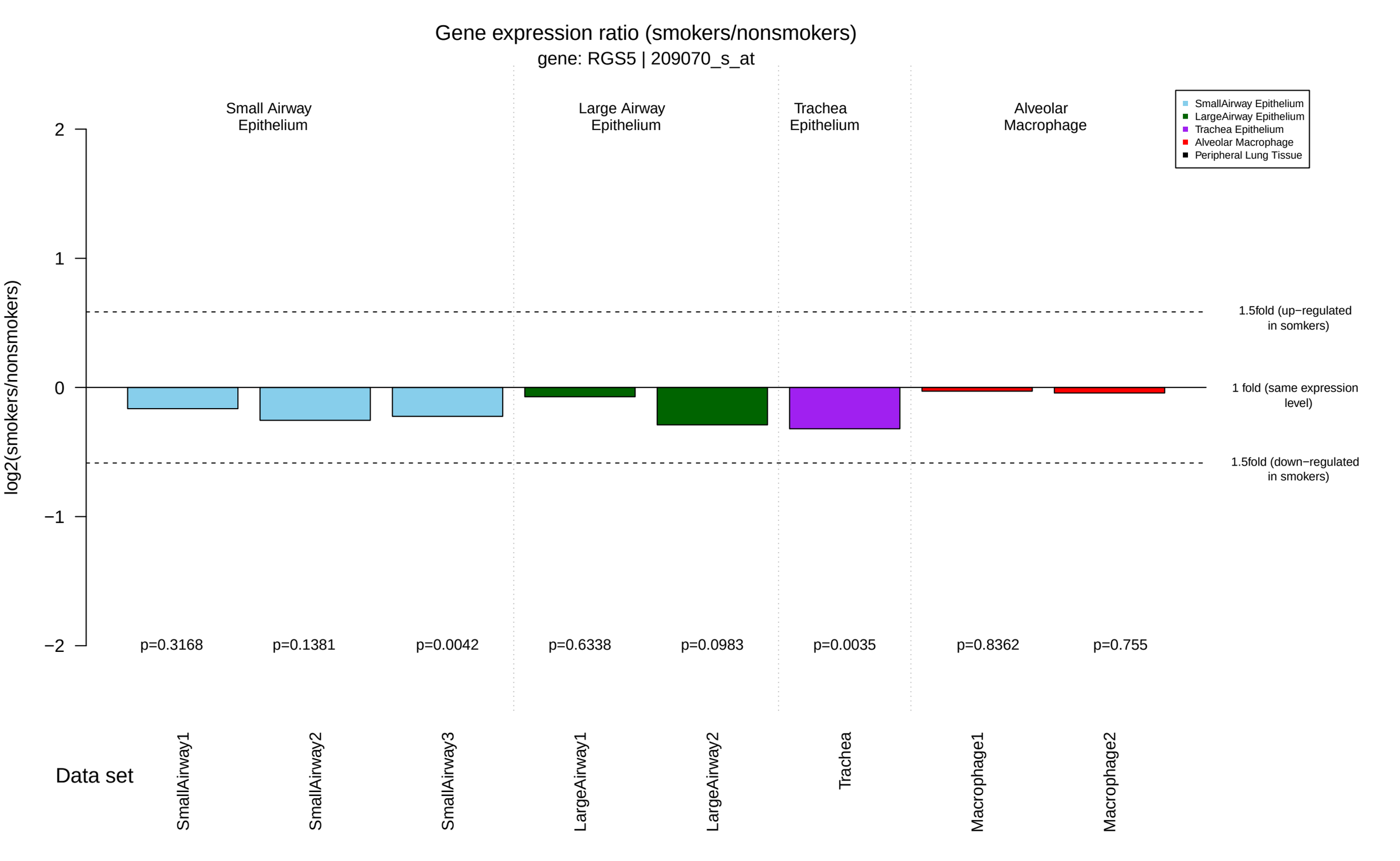
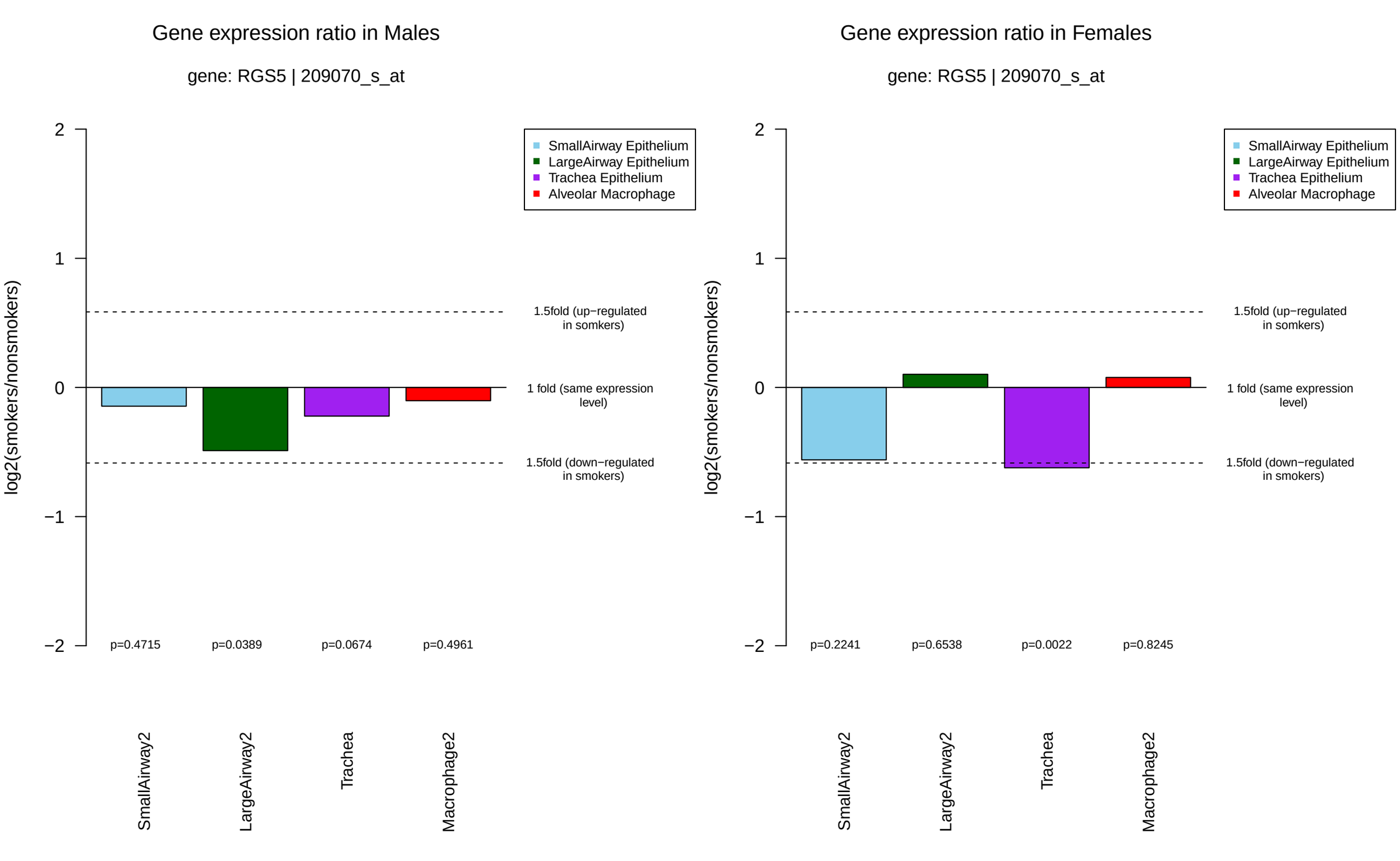
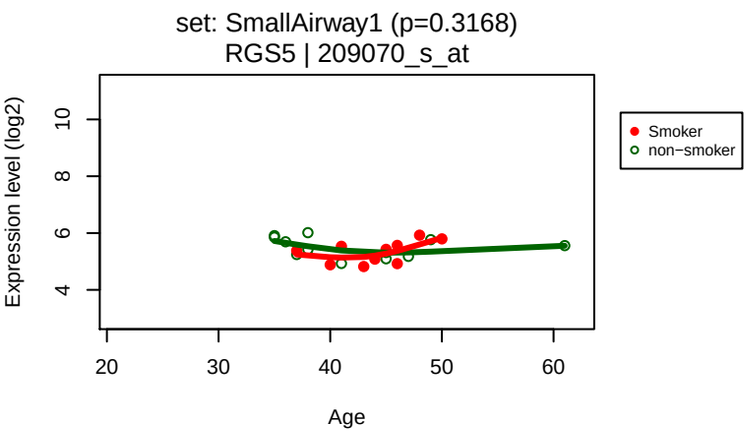
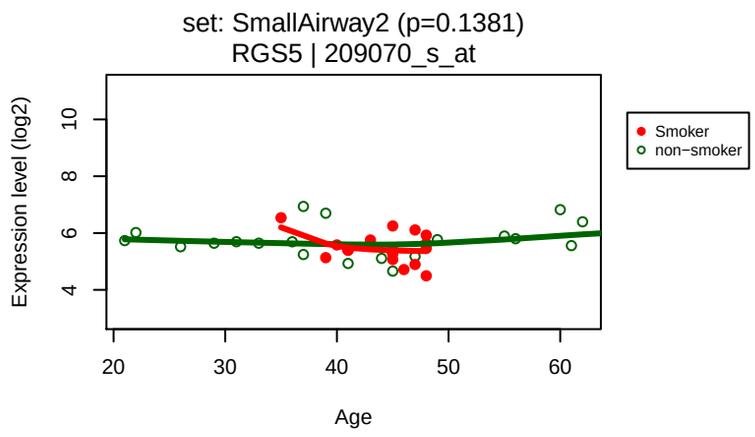
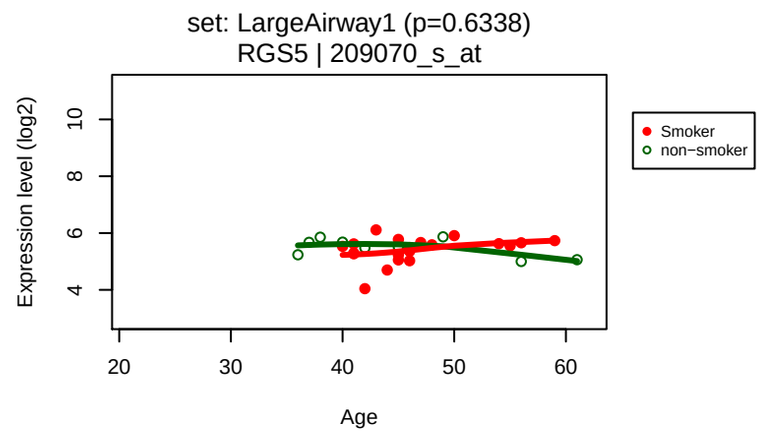

Gene expression of smokers and nonsmokers
gene: RGS5 | 209070_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3168
p=0.1381
p=0.0042
p=0.6338
p=0.0983
p=0.0035
p=0.8362
p=0.755
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

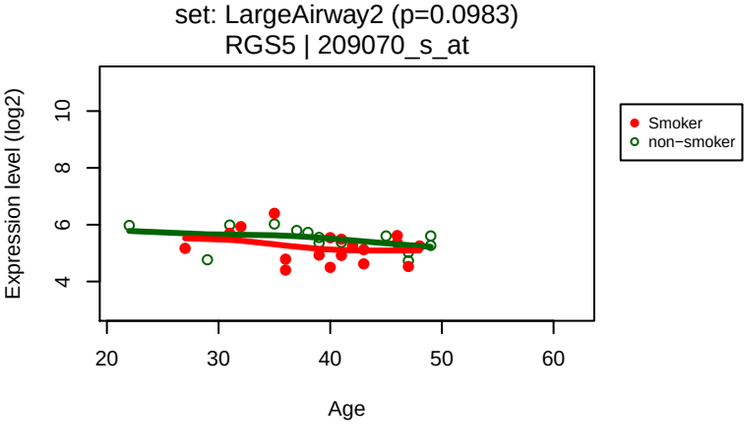

Gene expression ratio (smokers/nonsmokers)
gene: RGS5 | 209070_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3168
p=0.1381
p=0.0042
p=0.6338
p=0.0983
p=0.0035
p=0.8362
p=0.755
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

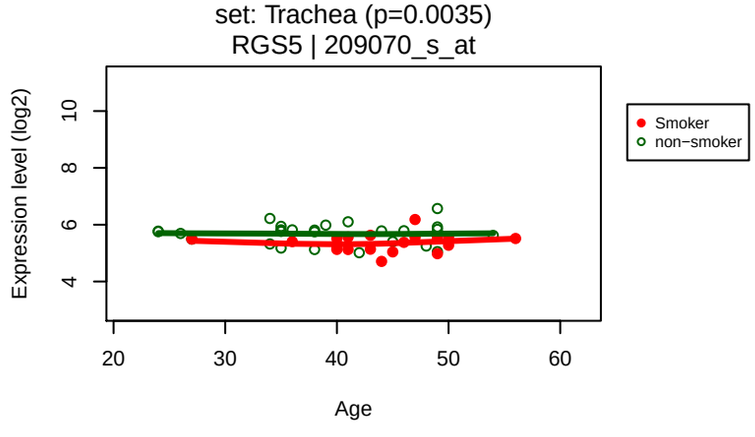

Gene expression ratio in Males
gene: RGS5 | 209070_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4715
p=0.0389
p=0.0674
p=0.4961
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS5 | 209070_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2241
p=0.6538
p=0.0022
p=0.8245
SmallAirway2
LargeAirway2
Trachea
Macrophage2

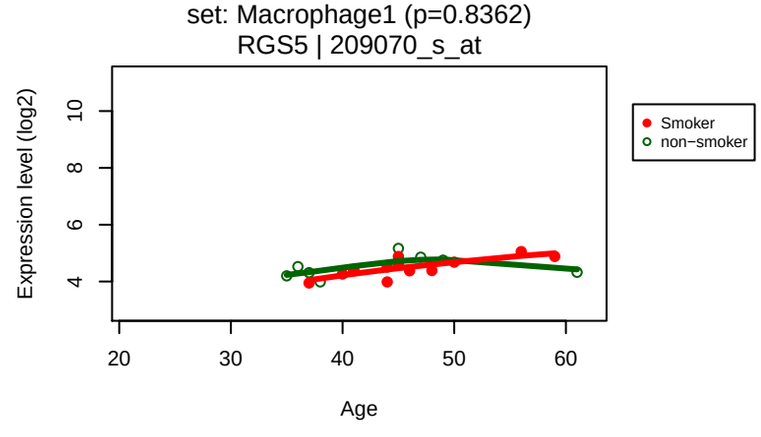

set: SmallAirway1 (p=0.3168)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker

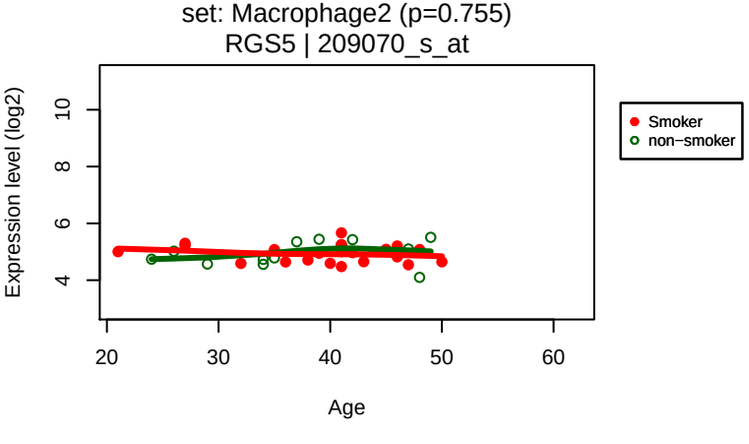

set: SmallAirway2 (p=0.1381)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker

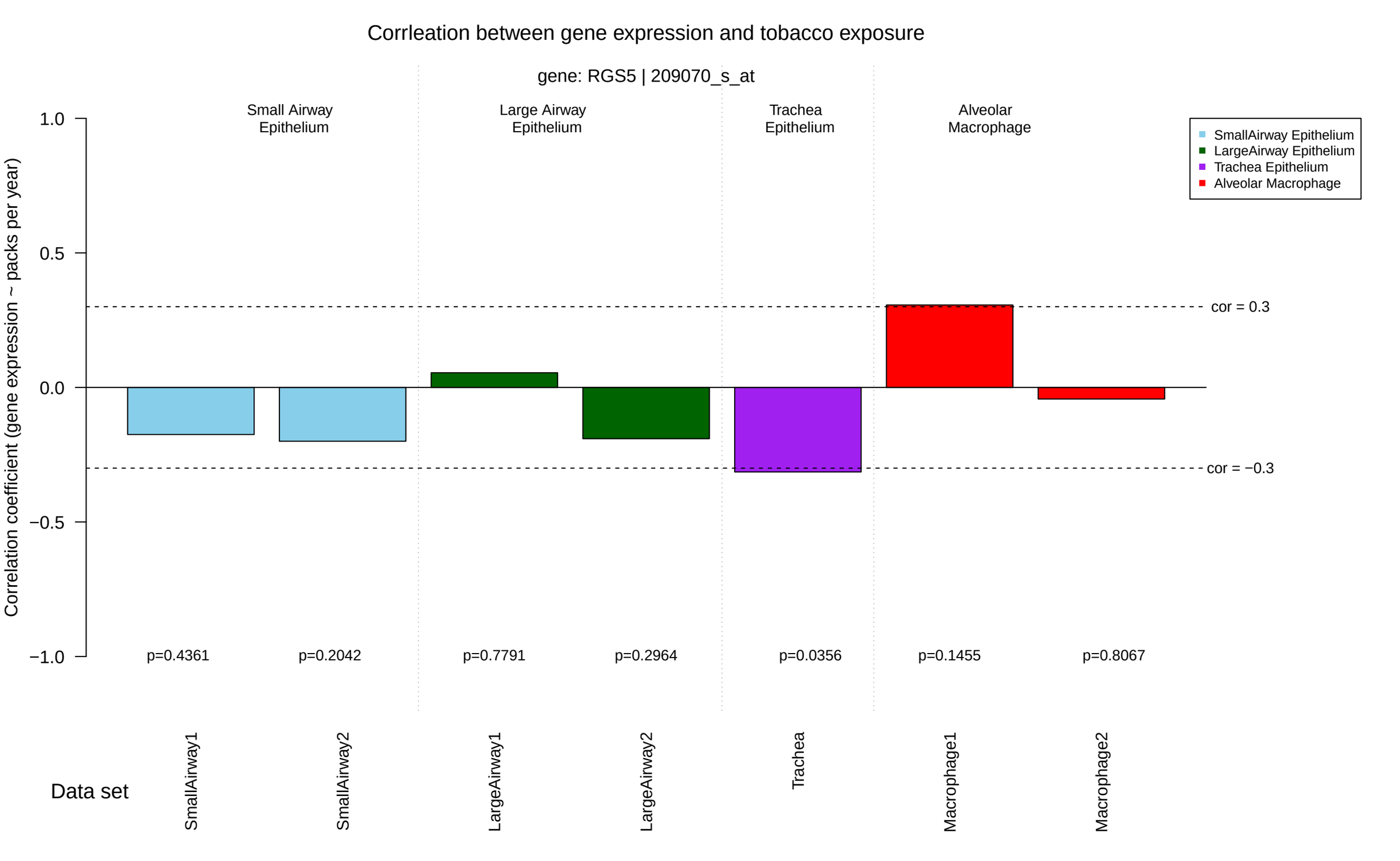

set: LargeAirway1 (p=0.6338)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0983)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0035)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8362)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.755)
RGS5 | 209070_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS5 | 209070_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4361
p=0.2042
p=0.7791
p=0.2964
p=0.0356
p=0.1455
p=0.8067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2